

Hierarchy-Aware Hyperbolic and Semantic Reranking for Ontology-Based Phenotype Linking

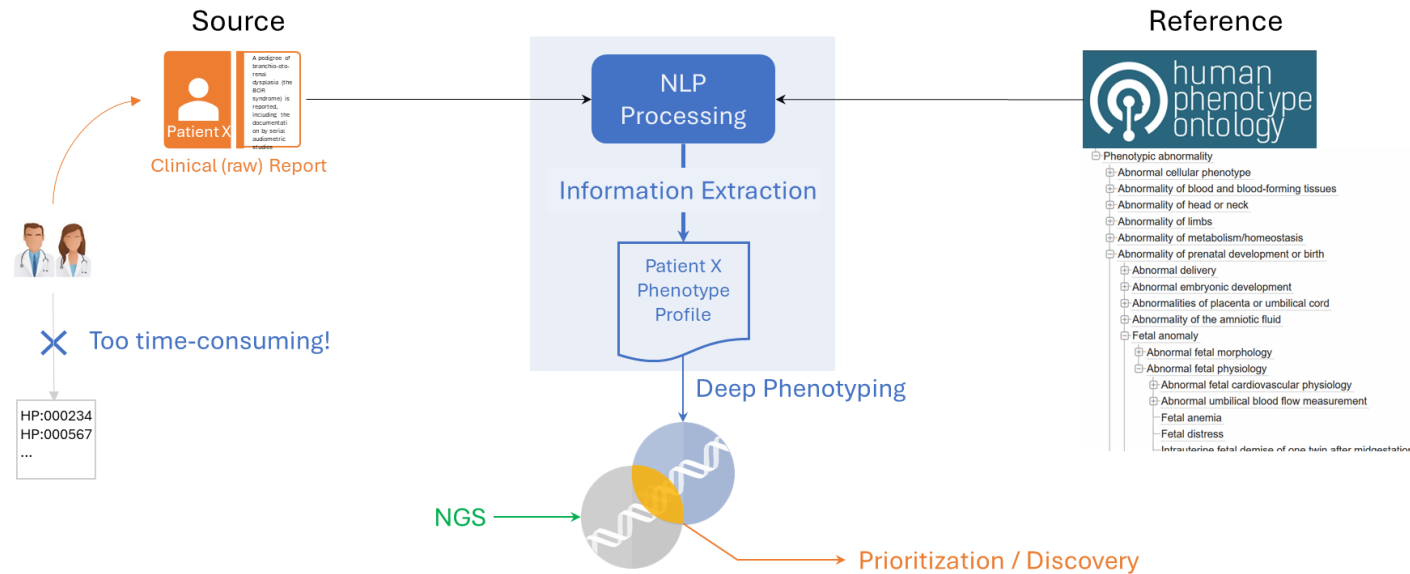


BioNLP Workshop

Thomas Labbé, Moussa Baddour, Axel Bonesteven, Paul Rollier, Marie de Tayrac, Olivier Dameron



Motivation: from reports to deep phenotyping



Diagnostic goal

Improve rare genetic disease diagnosis by recovering the most exhaustive phenotype profile associated with a patient.

Deep phenotyping requires an exhaustive, standardized HPO phenotype profile

NLP bottleneck

Clinical reports contain phenotype references, but with implicit and heterogeneous formulations

“difficulty pronouncing long words”
→ Expressive language delay

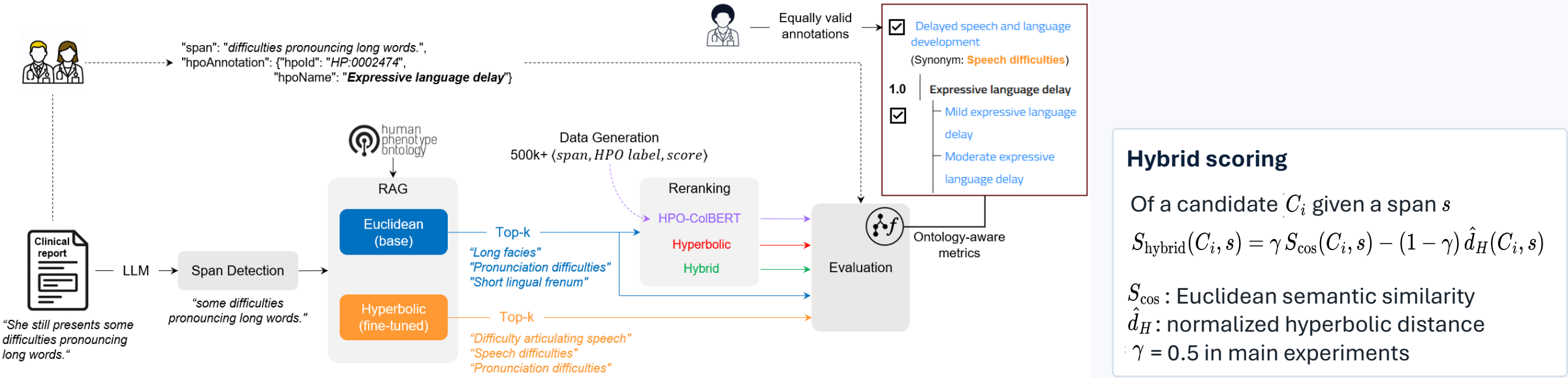
Modeling implication

The target space is not flat: HPO is hierarchical.

Candidates should be semantically relevant + close in the ontology + coherent

Research question: Can hyperbolic geometry help produce more accurate and ontology-coherent HPO candidate lists?

Approach: retrieve semantically, rerank with hierarchy



Semantic signal

Hierarchical signal

Clinical output

Hierarchy-aware evaluation

local lexical and contextual similarity

global HPO branch consistency

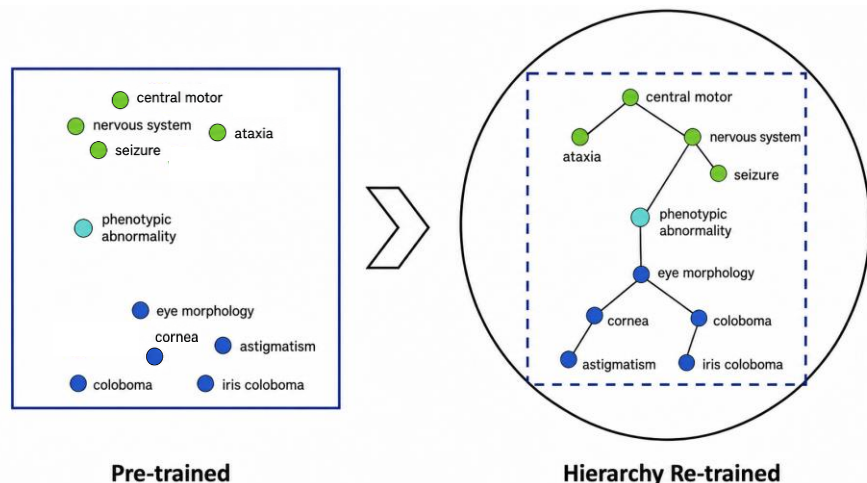
a ranked candidate list to review

Hypothesis: Hyperbolic geometry should not replace semantic retrieval: it should act as a structural prior that improves candidate-list coherence when HPO hierarchy matters.

Hyperbolic training: HPO as a hierarchy supervision signal

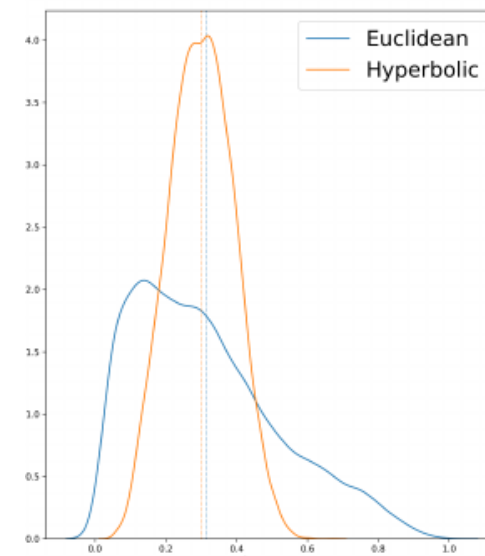
Training setup

- Base encoder: all-MiniLM-L12-v2
- Framework: Hierarchy Transformers
- Supervision: HPO child-parent triplets
- Synonym augmentation + negative sampling
- Loss: clustering + centripetal components

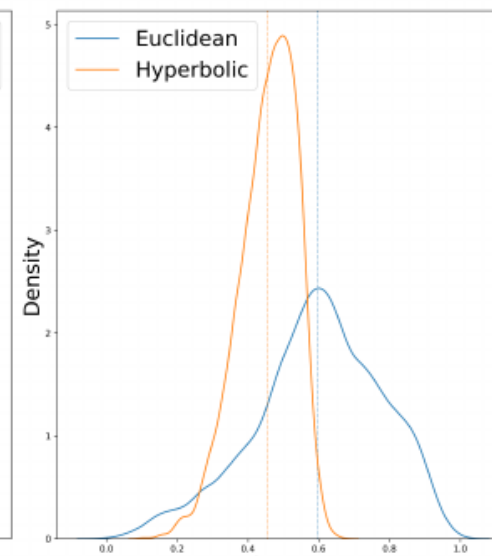


Adapted from He et al. 2024 - Hierarchy Transformers

Distribution of One Hop Distances



Distribution of Multi Hop Distances



Distance distributions: one-hop and multi-hop HPO pairs

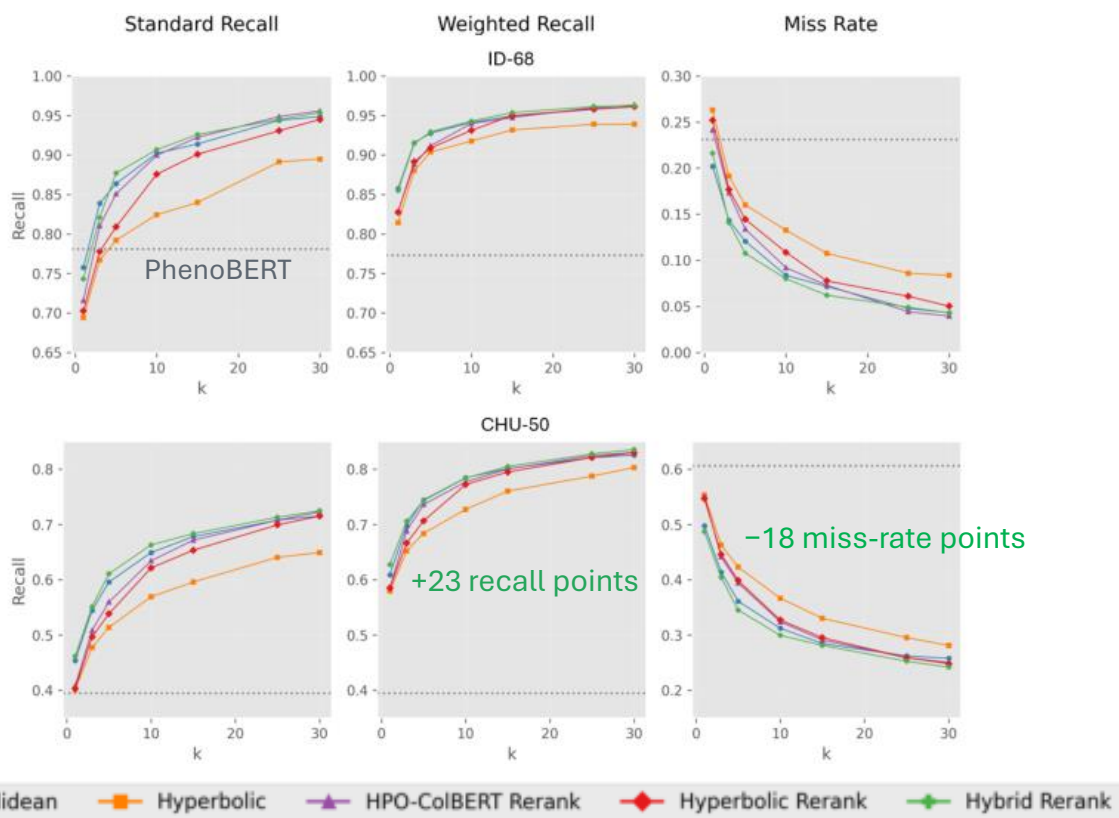
Interpretation

Narrower hyperbolic distributions indicate more faithful hierarchical organization, especially for multi-hop relations.

Results: hybrid reranking is strongest on real clinical notes

Evaluation framing Clinical priority: candidate-list quality, not only exact Top-1

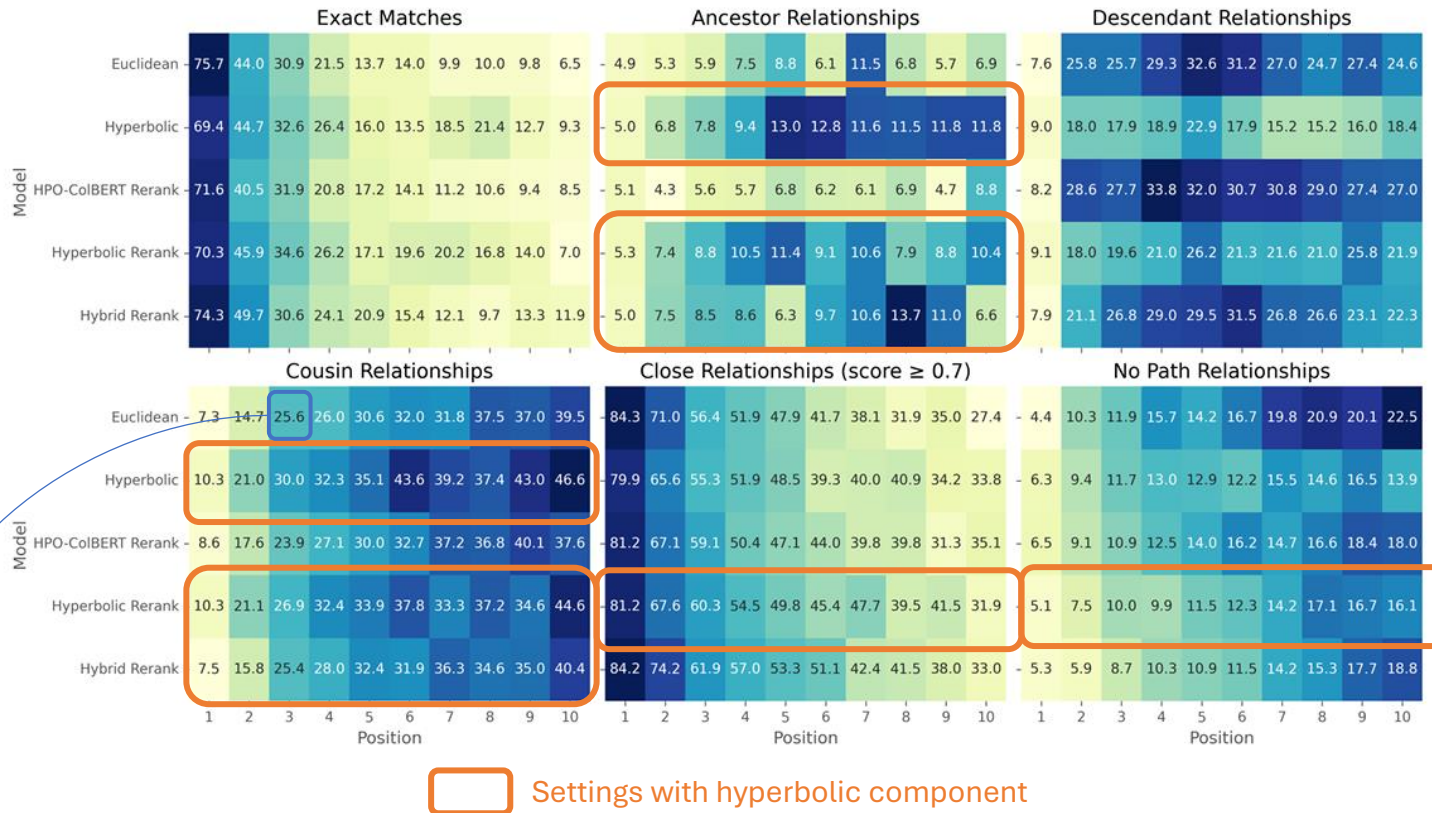
- Primary metric: **Recall@k** and **Miss rate@k**
- Secondary metrics: MRR/NDCCG
- Ontology metrics: average hop@k and branch coverage@k
- Exact match is too strict: close HPO relatives can be clinically meaningful.
- **Hierarchy-aware variants** turn ontology proximity into graded relevance.



Observations

- Hyperbolic-only retrieval underperforms: hierarchy is complementary, not sufficient
- ID-68 (explicit dataset): hybrid remains comparable to strong Euclidean retrieval, so hierarchy does not degrade strong explicit cases.
- CHU-50 (new curated dataset with ~30% implicit annotations) : hybrid improves weighted Recall@1 and weighted MRR
- **All retrieval-based methods outperform PhenoBERT from k = 1 on CHU-50**

Hyperbolic signals make candidate lists more ontology-coherent



Distribution of relationship types by position and settings for Top-10 candidates

25.6% of Top-3 candidates are cousins of the target phenotype for Euclidean settings

Ontology behavior

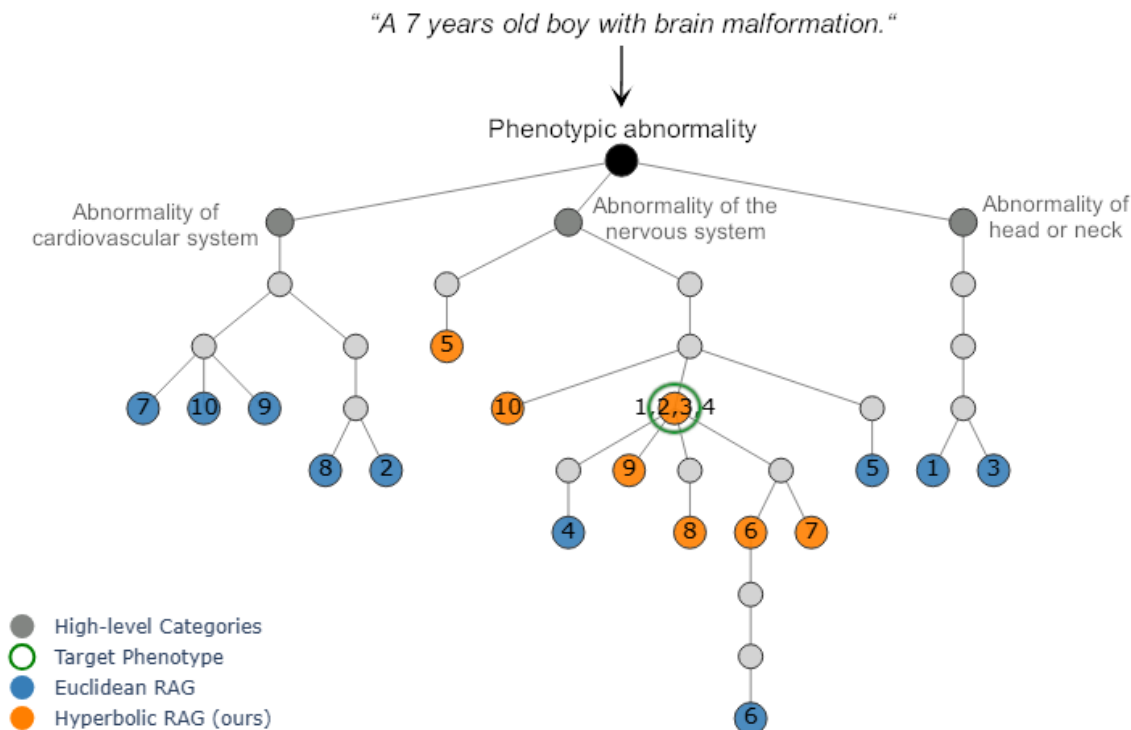
- Hyperbolic models show more ancestor and cousin relationships
- Fewer unrelated candidates at higher ranks
- Near misses become clinically meaningful HPO relatives, not arbitrary errors.

Interpretation

Hybrid keeps exact matching near the top while improving structural coherence

Takeaway: from clinical narratives to structured phenotype evidence

A **high-recall, ontology-coherent** shortlist is a better interface between NLP and clinical deep phenotyping.



Provided resources

- ✓ CHU-50 evaluation dataset in French and English
- ✓ Synthetic training set for the ColBERT reranker
- ✓ HPO hierarchy training set for the hyperbolic model
- ✓ Trained HPO hyperbolic embedding model
- ✓ Code for retrieval, reranking, and hierarchy-aware evaluation



github.com/labbeth/HyperRAG



[labbeth/HiT-all-MiniLM-L12-v2-hpo](https://labbeth.github.io/HiT-all-MiniLM-L12-v2-hpo)

Key references

Robinson et al. 2008 - Human Phenotype Ontology
Nickel & Kiela 2017 - Poincare embeddings
He et al. 2024 - Hierarchy Transformers
Santhanam et al. 2021 - ColBERTv2
Feng et al. 2023 - PhenoBERT baseline

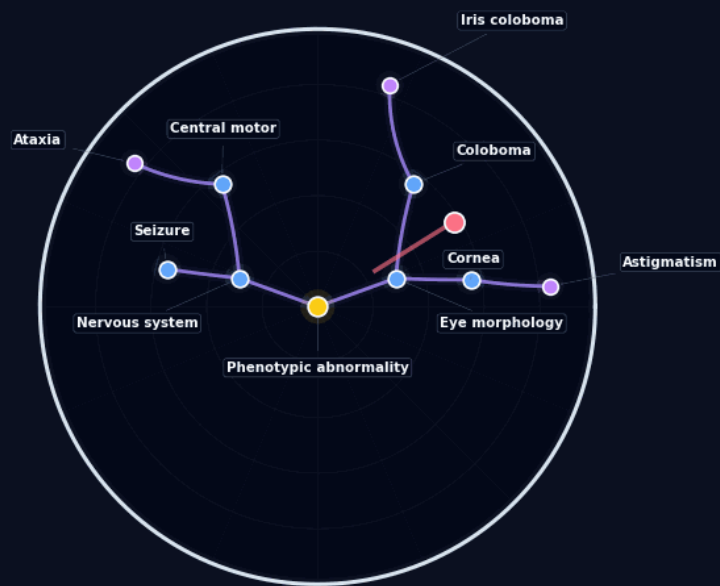
Euclidean embeddings capture **local semantics**. Hyperbolic embeddings capture global **HPO structure**. **Hybrid** reranking turns this complementarity into **structured evidence** for deep phenotyping.

Thanks

thomas.labbe@gmail.com

Hyperbolic Representation of Human Phenotype Ontology

Radial motion in the Poincaré disk



Hierarchical orbits around the root phenotype



● Radial motion ● Inner orbit ● Outer orbit

Phenotypes become **increasingly specific** toward the **Poincaré disk boundary**.
All colored dots move with the **same constant speed**; near the boundary, they **appear slower**.

Evaluation: candidate-list quality, not only exact Top-1

Primary objective

Recover relevant HPO candidates in the Top-k list so clinicians can review, refine, and validate a phenotype profile.

Primary metrics Recall@k • Miss rate@k

Secondary: MRR and NDCG for ranking quality

Ontology metrics: average hop@k and branch coverage@k

Clinical rationale: a consistent list reduces false negatives and helps final diagnosis in a deep phenotyping workflow.

Hierarchy-aware evaluation

Exact match is too strict: ancestors, descendants, and cousins can still be clinically meaningful.

$$w_{\text{direct}}(C, T) = \begin{cases} \frac{\alpha}{p(1 + |d|)}, & p > 0 \\ 1, & p = 0 \end{cases}$$

$$w_{\text{indirect}}(C, T) = \frac{\beta}{c(1 + d_l)}$$

p: number of ancestor / descendant steps

d: distance between candidate and target

c: descendants of the most specific common ancestor

d_l: distance to farthest HPO leaf

These weights turn ontology proximity into graded relevance for recall, MRR, and NDCG.

Hyperbolic model training set

Input ontology

HPO OWL file
+ synonyms
+ parent–child structure

Parsed with DeepOnto
and the ELK reasoner.

Objective

Pull child–parent terms together
and push unrelated terms apart in
hyperbolic space.

Triplet construction

Positive: ⟨child, parent, 1⟩
Negative: ⟨child, non-parent, 0⟩
Strategy: random negative sampling

Examples

Positive
⟨“Iris coloboma”, “Coloboma”, 1⟩

Negative
⟨“Iris coloboma”, “Abnormality of the
cardiovascular system”, 0⟩

Harder negative (same broad branch)
⟨“Iris coloboma”, “Abnormal retinal
morphology”, 0⟩

Synonym combinations are added,
then filtered to limit each synonym to
max. five occurrences.

Model training

Framework: Hierarchy Transformers
Base: all-MiniLM-L12-v2
Epochs: 20
Batch: 32 train / 64 eval
LR: 1e-5
Losses: hyperbolic clustering +
centripetal

Released on HF as

**labbeth/HiT-all-MiniLM-
L12-v2-hpo**

ColBERT reranker training set

1. Generate sentences

ChatGPT-4o-mini generated 10 clinical report sentences for each HPO term, with diversity constraints: implicit mentions, measurements, and varied formulations.

Example generated sentence

“She still presents some difficulties pronouncing long words.”

2. Extract + filter spans

The same LLM extracted the most precise span. Clinicians helped define heuristics to remove vague or uninformative spans.

Extracted span

“difficulties pronouncing long words”

Output: 91,760 unique high-quality spans

3. Build reranking triplets

Positive span–HPO pairs are scored 1. Negatives combine hard cases from the same branch and easy cases outside the branch; scores use hyperbolic distance to the target.

Concrete examples

Positive:

⟨span, “Expressive language delay”, 1⟩

Hard negative:

⟨span, “Speech articulation difficulties”, distance-based score⟩

Easy negative:

⟨span, “Abnormality of the cardiovascular system”, low score⟩

Final set: 510,371 triplets ⟨span, HPO label, score⟩